

The following resources and documentation provide the essential tools, frameworks, libraries, and learning materials required for the development of FinRelief AI. These references support environment setup, frontend and backend development, database management, AI integration, authentication, and overall project implementation.

Python 3.11+: Core programming language used to build the application backend, manage APIs, financial processing logic, and AI-powered negotiation modules.

<https://www.python.org/downloads/>

FastAPI Documentation: High-performance Python web framework used to create REST APIs for frontend-backend communication, authentication, financial processing, and AI integration services.

<https://fastapi.tiangolo.com/>

React.js Documentation: Frontend JavaScript library used to build the responsive borrower dashboard and interactive user interface components.

<https://react.dev/>

Vite Documentation: Modern frontend build tool used for fast React.js development, hot reload functionality, and optimized production builds.

<https://vitejs.dev/>

Google Gemini API: AI service used for generating intelligent negotiation strategies, settlement recommendations, and lender-specific negotiation letters.

<https://aistudio.google.com/app/apikey>

SQLAlchemy ORM Documentation: Python ORM library used for database management, model creation, and relational data handling within the platform.

<https://www.sqlalchemy.org/>

SQLite Documentation: Lightweight relational database used to securely store borrower information, loan details, settlement records, and AI history data.

<https://www.sqlite.org/docs.html>

Git Version Control: Source code management system used for tracking project changes, version history, and collaborative development workflows.

<https://git-scm.com/doc>

GitHub Repository Hosting: Cloud-based repository platform used for project hosting, version control, and source code management.

<https://github.com/>

Python Virtual Environment (venv): Environment isolation tool used to manage Python dependencies separately and maintain stable package versions across development setups.

<https://docs.python.org/3/library/venv.html>

Visual Studio Code: Recommended development environment used for coding, debugging, API testing, and project management.

<https://code.visualstudio.com/>

JWT Authentication Documentation: Secure authentication mechanism used for protected API access and user session management.

<https://jwt.io/>